VALEMONT OS · SETUP

The plan and tutorial for every connection

French version: docs/SETUP_GUIDE.md

For Théo and Pablo. This guide tells you **in what order** to connect each service, **where to click**, **what to paste** and **how to check** that it works. It is based on the project documents (they are the reference): `.env.example`, `docs/DEPLOYMENT.md`, `docs/INTEGRATIONS.md`, `docs/INBOX.md`, `docs/TRACKER.md`, `docs/SECURITY.md`, `docs/ACCEPTANCE.md`, `docs/DECISIONS.md` (D086, D087). The steps at Google, Vercel, Wise, Stripe, Meta and LinkedIn were checked against their official documentation in November 2026 (links in each section). These interfaces change often: if a menu has a different name, search for the keyword given.

The app's interface is in French (Théo) and Spanish (Pablo); this guide quotes the French labels with the English meaning in parentheses.

Three rules before you start

1. **A key, token or password is pasted in one of two places only**: in the application (**Réglages › Intégrations › Configurer** (Settings › Integrations › Configure), where it is encrypted), or, for the few base settings, in **Vercel › Settings › Environment Variables**. Never in a chat (including with Claude), never in an email, never directly in the Google Sheet, never in the code.
2. **Every account belongs to Valemont** (your Google addresses, your Meta Business Manager, your Vercel account). The Claude account that built the application is never used on your real resources.
3. **What you save in the application works immediately**: no redeploy. Only a change in Vercel needs **Vercel › Deployments › ⋯ › Redeploy**.

In the application, everything is controlled from **Réglages › Intégrations**: one row per service (16 in total), with its status (*Connecté* (Connected), *Non configuré* (Not configured), *Erreur* (Error), *À renouveler* (Needs renewal), *Import uniquement* (Import only), *En attente d'approbation* (Awaiting approval), *Démo, simulé* (Demo, simulated)), the **names** of the values (« défini dans Vercel » (set in Vercel), « enregistré dans l'app » (saved in the app) or missing, never the value itself) and the buttons **Configurer** (Configure), **Tester la connexion** (Test connection), **Synchroniser** (Sync), **Autoriser / Autoriser à nouveau** (Authorize / Authorize again) where they make sense. At the bottom, **Réglages de base (définis dans Vercel)** (Base settings, set in Vercel) lists, read-only, what stays in Vercel.

1. The plan

Tick the boxes as you go. "Who" is a suggestion: what matters is that the person has admin rights on the account concerned.

Connecting a service is always the same (Phases 4 to 9)

1. **Réglages** › **Intégrations** › the service's row › **Configurer** (Configure).
2. Follow the numbered steps (each has a link to the right page at the provider; the addresses to copy there are shown with a Copy button).
3. **Paste** what the provider gives you, or click **Générer** (Generate) for a secret we create ourselves: it is shown **only once**, copy it straight away into the other service.
4. **Enregistrer et tester** (Save and test): green = « Ça marche. » (It works.), red = the reason in plain words. Nothing to redeploy.

A variable with the same name in Vercel is still possible (the old way): **a variable in Vercel takes priority**, and the form then shows it as « défini dans Vercel » (set in Vercel) and cannot overwrite it.

Phase 0 · Prerequisites and accounts (both of you · 30 min)

- ☐ Check that you have: a Google account each (ideally Google Workspace, same domain), a GitHub account, a Vercel account, admin access to Wise Business, Stripe, Meta Business (Facebook Page + Instagram professional accounts), the LinkedIn Pages, GA4, Search Console, the website hosting (Netlify or other), and Make, Zapier or n8n if you want the message relay.
- ☐ Nothing to install: no terminal, no Node.js. Everything happens in the browser.
- ☐ A shared password manager (1Password, Bitwarden…) to keep a copy of the base settings.

Unlocks: nothing yet, but it saves you from getting stuck halfway.

Phase 1 · Google: project, Sheet, sign-in (Théo, or whoever holds the Google admin account · 1 h 30)

- ☐ Create the Google Cloud project and enable the 4 APIs (Sheets, Drive, Calendar, Gmail) (§2.2).
- ☐ Create the **service account** and download its JSON key (§2.2).

- [ ] Create the empty main Google Sheet and share it with the service account (§2.3).
- [ ] (Optional) Each of you creates a private Sheet and shares it (§2.3).
- [ ] Share the "Valemont OS" Drive folder (Viewer), prepare the backup folder, share the calendars (§2.5).
- [ ] OAuth consent screen + OAuth client (§2.4); the exact redirect addresses come in Phase 2.

Unlocks: the database (Sheet), Google sign-in for both founders, Files, Calendar.

Phase 2 · Going live (Théo or Pablo · 30 min)

- [ ] Click the README's **Deploy with Vercel** button and fill in the form with **the base settings only**: `APP_MODE` = `live`, `SESSION_SECRET`, your two Google addresses, the OAuth client ID and secret, the Sheet ID, the service account JSON key (pasted as-is). **Pro** plan, Paris region (§2.7).
- [ ] Open `https://<your-app>/setup`: the page says what is still missing (names only, never a value) and shows the 3 redirect addresses to paste in Google Cloud (§2.4).
- [ ] On `/setup`, click « Générer un secret » (Generate a secret) to create `CONNECTORS_ENCRYPTION_KEY` (the key that encrypts what you will paste into the app) and, with another click, `CRON_SECRET`; add them in Vercel, keep a copy in the password manager, **Redeploy** once, then « Vérifier à nouveau » (Check again).

Unlocks: the application online in LIVE mode (without the base settings, it shows a configuration page, never the demo). `APP_BASE_URL` sets itself on Vercel.

Phase 3 · First start (both of you · 10 min)

- [ ] Sign in with Google: you land on the **Installation** page › « Préparer la base de données » (Prepare the database). The app creates the Sheet's tabs and headers, then checks everything. Then My Day opens (§2.8).
- [ ] Théo signs in, Pablo signs in from another browser; a non-founder account is refused.
- [ ] Réglages › Calendriers (Settings › Calendars) and Réglages › Espace de travail (Settings › Workspace): paste the calendar IDs and the Drive folder ID.

Unlocks: the whole core application (Ma journée (My Day), CRM, Appels (Calls), Tâches (Tasks), Agenda (Calendar), Fichiers (Files)).

Phase 4 · Money: Wise and Stripe (whoever administers the bank · 45 min)

- [ ] Wise › **Configurer**: read-only token (+ optional webhook) › **Enregistrer et tester** (§2.9).
- [ ] Stripe › **Configurer**: read-only restricted key + webhook secret › **Enregistrer et tester** (§2.10).

Unlocks: Finances (balances, transactions, sales, fees, payouts), MRR in Finances and the Command Center, USD→EUR rate for supplier payments in dollars. The application can never move money.

Phase 5 · Email (Gmail) and calendar (each for yourself · 10 min each, + possible Google delay)

- [ ] Check that `gmail.send` is on the consent screen, then each of you clicks **Autoriser** (Authorize) on the Gmail row.
- [ ] Check that the Agenda merges the calendars properly (the other founder's personal calendar shows as « Occupé » (Busy)).

Unlocks: one-click email sending from Appels and Croissance › Messages (Growth › Messages) (always after your click). ⏳ With an "External" Google app, consent is requested again often, and publishing to production requires **Google verification** (see §2.6).

Phase 6 · Website and messages to handle (Pablo or whoever manages the websites · 1 h)

- [ ] Formulaires du site (Website forms) › **Configurer** › **Générer** › copy the secret to the website host (§2.11).
- [ ] Messages à traiter (To reply inbox) › **Configurer** › **Générer** › copy the secret into Make / Zapier / n8n (§2.12).

Unlocks: every website form becomes an inbound lead + an « À répondre » (To answer) item; comments and messages arrive in Ma journée.

Phase 7 · Social: Instagram, LinkedIn, GA4, Search Console (Théo or Pablo · 2 h + Meta/LinkedIn delays)

- [ ] Instagram › **Configurer**: token (system user) + account ID › **Enregistrer et tester** (§2.13).

- [ ] Instagram: comments/messages webhook → Messages à traiter (§2.12 B). ⏳ **Meta App Review** for accounts that have no role in the app.
- [ ] LinkedIn: export imports (.xls Page, .xlsx creator). ⏳ Community Management API: optional **access request**, long and not guaranteed.
- [ ] GA4 and Search Console: CSV exports imported every week.

Unlocks: Croissance (Growth: channels, costs, funnels), Studio contenu (Content Studio: numbers per post), Croissance › SEO.

Phase 8 · Personal tracker (each for yourself · 15 min)

- [ ] Mon tracker (My tracker) › **Configurer**: paste your tracker's address, **Générer** the secret (and the feed token), copy them into your tracker › **Enregistrer et tester** (§2.16).

Unlocks: each of your activities sent to your personal tracker, and a feed it can read.

Phase 9 · Backups and restore test (Théo · 30 min)

- [ ] Sauvegardes (Backups) › **Configurer**: paste the backup Drive folder's address › **Enregistrer et tester** (§2.17).
- [ ] (Optional) Capture planifiée (Snapshot job) › **Configurer**, only if a scheduler other than Vercel calls it.
- [ ] The next day: Réglages › Diagnostic › Sauvegardes › « Vérifier » (Verify) on the day's copy (restore drill).

Unlocks: verified daily copy of the Sheet, Instagram snapshot every 6 h, a plan B if something goes wrong.

Phase 10 · Final checks (both of you · 45 min)

- [ ] Réglages › Intégrations: every configured row is **Connecté** after **Tester la connexion**; GA4 and Search Console = *Import uniquement* (normal); LinkedIn = *En attente d'approbation* (normal).
- [ ] Réglages › Diagnostic (Settings › Diagnostics): health « prête » (ready), backups « vérifiée » (verified).
- [ ] Go through the "Stage 4 gate: live validation" list in `docs/ACCEPTANCE.md` and record the date of each box in `docs/STATUS.md` (nothing is VERIFIED_LIVE before that).

2. The tutorial

2.1 Secrets: who creates them, where they go

You no longer need a terminal to make a secret:

Secret	How to get it	Where it goes
`SESSION_SECRET`	Your password manager's generator (48+ characters), or the « Générer un secret » (Generate a secret) button on the `/setup` page	Vercel (Deploy button form)
`CONNECTORS_ENCRYPTION_KEY`, `CRON_SECRET`	« Générer un secret » on `/setup` (one click per variable, a different value each time; made in your browser, sent nowhere)	Vercel + a copy in the password manager
Service secrets (messages, website, tracker, jobs)	**Générer** (Generate) button in **Configurer**: made on the server, saved encrypted, shown **only once**	Copy it straight away into the other service (relay, website host, tracker, scheduler)
Keys given by a provider (Wise, Stripe, Meta)	At the provider (steps in each section)	Paste them in **Configurer**

One secret per use: never reuse the same one for two variables. Keep `CONNECTORS_ENCRYPTION_KEY` in the password manager: without it, the keys saved in the app become unreadable (you would have to paste them again).

Advanced (optional): in a terminal, `openssl rand -base64 48` (`SESSION_SECRET`), `openssl rand -base64 32` (`CONNECTORS_ENCRYPTION_KEY`), `openssl rand -hex 32` (other secrets) give the same result.

2.2 Google Cloud: project, APIs, service account

What it's for: the service account is the "robot" that reads and writes the Google Sheet (the only database), reads the shared Drive folder, reads the calendars and makes the backups.

What you need: your Google account (preferably Workspace), 30 minutes.

Steps

1. Go to https://console.cloud.google.com › project picker › **New Project** › name "Valemont OS". (Docs: https://developers.google.com/workspace/guides/create-project)

2. **APIs & Services › Library**: enable **Google Sheets API**, **Google Drive API**, **Google Calendar API**, **Gmail API**. (Docs: https://developers.google.com/workspace/guides/enable-apis)
3. **IAM & Admin › Service Accounts › Create service account**: name `valemont-os`, **no role** (its rights come only from the shares you make). Note its address: `valemont-os@<project>.iam.gserviceaccount.com`. (Docs: https://cloud.google.com/iam/docs/service-accounts-create)
4. Open the service account › **Keys** tab › **Add key › Create new key › JSON**. A file downloads. (Docs: https://cloud.google.com/iam/docs/keys-create-delete)
5. Open the file with a text editor (Notepad, TextEdit), copy **all** of its content.
6. Paste it as-is into the Deploy button form (Phase 2) as `GOOGLE_SERVICE_ACCOUNT_KEY_BASE64` (the file in base64 also works), then **delete the file** (and empty the trash).

Where it goes: Vercel (base setting): `GOOGLE_SERVICE_ACCOUNT_KEY_BASE64`.

Common problems

- *"Service account key creation is disabled"*: recent Google Cloud organizations block keys by default (policy `iam.disableServiceAccountKeyCreation`). The Workspace admin must lift this policy **for this project only** (IAM › Organization policies), create the key, then put the policy back.
- Lost or leaked key: create a new key, replace the variable in Vercel, Redeploy, **then** delete the old key.

2.3 Google Sheets (database) and private Sheets

What it's for: everything you enter is read and written here. The private Sheets hold each founder's private tasks and applications, invisible to the other.

Steps

1. On https://sheets.google.com, create an **empty** spreadsheet "Valemont OS" (the application never creates it).
2. **Share** › paste the service account's address › role **Editor** › untick "Notify people".
3. Copy the ID from the URL: `https://docs.google.com/spreadsheets/d/<THIS-ID>/edit` → `SHEETS_SPREADSHEET_ID`.
4. (Optional) **Each of you** creates, **from your own account**, a spreadsheet "Valemont OS – privé ", shares it with the service account as **Editor** (and with no one else) → `PRIVATE_SHEET_ID_THEO`, `PRIVATE_SHEET_ID_PABLO` (add them in Vercel, then Redeploy).
5. The tabs and headers are created in Phase 3 by the app's « Préparer la base de données » button.

Where it goes: Vercel (base settings): `SHEETS_SPREADSHEET_ID` (required), `PRIVATE_SHEET_ID_THEO`, `PRIVATE_SHEET_ID_PABLO` (optional).

Check: Réglages › Diagnostic › « Vérifier » (database check): everything « OK ».

Common problems

- *Permission denied*: the share was made with a different address, or as Viewer instead of Editor.
- Don't rename or delete tabs/columns by hand: the application finds columns by their name. Adding formatting rows or formulas in the Sheet breaks reading (Diagnostic reports it).
- The `IntegrationSecrets` tab holds only unreadable encrypted text: don't touch it (an edited or moved cell no longer decrypts, and the key would have to be pasted again in **Configurer**).

2.4 Google Sign-In (sign-in for both founders)

What it's for: only your two Google addresses can sign in. The same OAuth client is also used for the Drive (`drive.file`) and Gmail (`gmail.send`) permissions, requested later, one at a time.

Steps (the menu is now called **Google Auth Platform** in the console)

1. **Google Auth Platform › Branding**: name "Valemont OS", support email = a founder's address, authorized domain = your domain. (Docs: https://developers.google.com/workspace/guides/configure-oauth-consent)
2. **Audience**:
   - if you are **both in the same Google Workspace**: type **Internal** (the simplest: no verification, no test users);
   - otherwise: type **External**, status **Testing**, and add both your addresses under **Test users**.
3. **Data access › Add or remove scopes**: `openid` , `.../auth/userinfo.email` , `.../auth/userinfo.profile` , `.../auth/drive.file` , `.../auth/gmail.send` . (The last two are only requested when you use them: this is incremental authorization, https://developers.google.com/identity/protocols/oauth2/web-server#incrementalAuth)
4. **Clients › Create client** › type **Web application** › name "Valemont OS". **Authorized redirect URIs**: copy the three exact addresses shown on the app's `/setup` page (Phase 2). They look like this (no trailing `/`):
   - `https://<your-app>/api/auth/google/callback`
   - `https://<your-app>/api/auth/google/drive/callback`
   - `https://<your-app>/api/auth/google/gmail/callback` If you create the client before deploying, come back and add these addresses afterwards.
5. Copy the **Client ID** → `GOOGLE_OAUTH_CLIENT_ID` and the **Client secret** → `GOOGLE_OAUTH_CLIENT_SECRET` (straight into the Deploy button form).

Where it goes: Vercel (base settings): `GOOGLE_OAUTH_CLIENT_ID` , `GOOGLE_OAUTH_CLIENT_SECRET` , `FOUNDER_THEO_EMAIL` , `FOUNDER_PABLO_EMAIL` ; optional: `FOUNDER_THEO_TIMEZONE` (default `Europe/Paris`), `FOUNDER_PABLO_TIMEZONE` (default `Europe/Madrid`), `APP_BASE_URL` (automatic on Vercel).

Check: open the application, « Se connecter avec Google » (Sign in with Google): Théo and Pablo get in, a third address is refused.

Common problems

- `redirect_uri_mismatch` : the registered URI doesn't exactly match the app's address (https, no trailing `/` , same subdomain). If you change domain, add the 3 new URIs (the `/setup` page gives them).
- *"Access blocked: app has not completed the Google verification process"*: in External/Testing, the address is not among the test users.

2.5 Google Drive and Google Calendar

What it's for: Fichiers (Files) lists the "Valemont OS" folder; uploading a file uses **your** `drive.file` permission (the file belongs to you). Agenda (Calendar) merges the shared "Valemont" calendar and your calendars; the other founder's personal calendar only shows as « Occupé » (Busy). Founder meetings are written to the shared calendar.

Drive steps

1. In Google Drive, "Valemont OS" folder › **Share** › service account as **Viewer**.
2. Copy the ID from the URL `https://drive.google.com/drive/folders/<ID>` and paste it into the application: **Réglages › Espace de travail › Identifiant du dossier Drive** (Settings › Workspace › Drive folder ID).

Calendar steps (Docs: https://support.google.com/calendar/answer/37082)

1. Google Calendar › ⚙ **Settings** › choose the shared "Valemont" calendar › **Share with specific people or groups** › add the service account with **Make changes to events**.
2. Each of you, for your personal calendar: same path, but with **See only free/busy (hide details)**.
3. In each calendar › **Integrate calendar › Calendar ID**: copy it into **Réglages › Calendriers**.

Where it goes: in the application (Réglages › Espace de travail, Réglages › Calendriers); no variable.

Check: Réglages › Intégrations › Google Drive and Google Calendar › **Tester la connexion**; open Fichiers and Agenda. First file upload: Google asks you for the Drive permission once (it lasts about 1 h, then one click on **Autoriser à nouveau**).

Common problems: empty calendar = wrong ID or forgotten share; invisible file = it isn't in the shared folder.

2.6 Gmail (send only)

What it's for: the emails prepared by the application (after a call, follow-ups, Croissance › Messages) are sent from **your** Gmail **when you click**. The application never reads your inbox. Without Gmail connected, it opens a pre-filled Gmail draft: nothing is blocked.

Steps

1. Check (step 3 of §2.4) that the Gmail API is enabled, that `gmail.send` is in Data access and that the URI `/api/auth/google/gmail/callback` is registered.
2. Each of you, in the application: **Réglages › Intégrations › Gmail › Autoriser** (or « Connecter Gmail » (Connect Gmail) in Appels). Google shows the consent screen: accept.

Where it goes: nothing to paste (same OAuth client).

Check: the Gmail row shows the authorization as active in this browser. There is **no** « Tester la connexion » for Gmail (a test would send a real email): send yourself a test email from a contact record to your address.

⏳ **Google delays and limits** (Docs: https://developers.google.com/workspace/gmail/api/auth/scopes, https://support.google.com/cloud/answer/13463073)

- `gmail.send` is a **sensitive** scope.
- **Internal** app (Workspace): no verification, nothing to wait for. This is the recommended choice.
- **External app in Testing**: works for your two addresses ("unverified app" screen to accept), but Google asks for consent again often (at least every 7 days).
- Moving an **External app to production** with `gmail.send` requires **Google verification** (demo video, privacy policy, verified domain): allow **several days to a few weeks**.

2.7 Vercel: going live

What it's for: hosting the application, keeping the base settings, running the two scheduled jobs.

Steps

1. Click the **Deploy with Vercel** button at the top of `README.md` ("Deploy (LIVE, no terminal)" section). Vercel copies the repository into your GitHub account and shows a form that asks only for the base settings: `APP_MODE` (`live`), `SESSION_SECRET` (§2.1), `FOUNDER_THEO_EMAIL`, `FOUNDER_PABLO_EMAIL`, `GOOGLE_OAUTH_CLIENT_ID`, `GOOGLE_OAUTH_CLIENT_SECRET` (§2.4), `SHEETS_SPREADSHEET_ID` (§2.3), `GOOGLE_SERVICE_ACCOUNT_KEY_BASE64` (the JSON file pasted as-is, §2.2). Click **Deploy**. If the Vercel project already exists, add these variables in **Settings › Environment Variables** instead. (Docs: https://vercel.com/docs/getting-started-with-vercel/import)
2. **Pro plan**: the Hobby plan is restricted to personal non-commercial use and doesn't allow a job every 6 h. A single paying member is enough. (Docs: https://vercel.com/docs/limits/fair-use-guidelines, https://vercel.com/docs/cron-jobs/usage-and-pricing)
3. **Settings › Functions**: region **Paris (cdg1)**; leave the Node.js runtime and Fluid Compute (default).

4. Open `https://<your-app>/setup`. For each base setting, the page says whether it is set, missing, to fix, automatic or default (never its value), and where to find it. It shows the three exact **redirect URIs** to paste in Google Auth Platform › Clients (§2.4).
5. Still on `/setup`, « Générer un secret » (Generate a secret) → add the value in Vercel as `CONNECTORS_ENCRYPTION_KEY`; « Générer un secret » a second time → `CRON_SECRET`. In **Settings › Environment Variables**, environment **Production**, tick **Sensitive** (the value then becomes unreadable, even for you) and keep a copy in the password manager. (Docs: https://vercel.com/docs/environment-variables)
6. **Deployments › ··· › Redeploy**, then « Vérifier à nouveau » (Check again) on `/setup`: everything should be green.
7. (Optional) **Settings › Domains** › add `os.<domain>` › create the DNS record shown (CNAME) at your registrar › wait for the certificate. `APP_BASE_URL` follows Vercel's production address automatically; you can also set it (`https://os.<domain>`, no trailing `/`, then Redeploy). Add the 3 new redirect URIs at Google. (Docs: https://vercel.com/docs/domains/working-with-domains/add-a-domain)
8. Scheduled jobs: nothing to create, `vercel.json` declares the daily backup (02:17 UTC) and the Instagram snapshot (every 6 h). Vercel calls them with `CRON_SECRET`. They only run on the production deployment. Visible in **Settings › Cron Jobs**. (Docs: https://vercel.com/docs/cron-jobs/manage-cron-jobs#securing-cron-jobs)

Where it goes: Vercel. This is the only step where you fill in Vercel; everything else happens in the app.

Never put in production: `SHEETS_API_BASE_URL`, `GOOGLE_API_BASE_URL`, `WISE_API_BASE_URL`, `STRIPE_API_BASE_URL`, `DEMO_BACKEND`, `FAKE_SHEETS_URL`, `FAKE_SHEETS_SPREADSHEET_ID`, `DEMO_ACCESS_CODE` (test and demo tools). `INSTAGRAM_API_BASE_URL` only in the Instagram Login case (§2.13).

Check: `https://<your-app>/api/health` answers `{"status":"ok"}` or `"degraded"` (`fail` = a base setting is missing or the Sheet is unreachable).

Common problems: « configuration requise » (configuration required) page (`/setup`) = a base setting is missing (its name is shown); variable added in Vercel but nothing changes = not redeployed.

2.8 First start (Phase 3)

No terminal: the database is prepared from the app.

1. Open the app and click « Se connecter avec Google » (Théo or Pablo, whoever starts).
2. While the Sheet is not ready, you land on the **Installation** page. Click « Préparer la base de données » (Prepare the database): the app creates tabs and headers, freezes headers, adds a warning-only protection and the starting settings, then runs the check. Progress and a per-tab result are shown. Safe to click again: it never deletes or overwrites anything.

3. If the service account lacks **Editor** access, the page says so and shows the exact service account address: Share › paste the address › Editor › untick notification › Share, then click again.
4. Your private Sheet (if `PRIVATE_SHEET_ID_<name>` is set) is prepared at the same time; each founder prepares their own at first sign-in (the app never touches the other founder's private Sheet).
5. Then My Day opens. Pablo signs in from another browser; a non-founder account is refused. Fill in Réglages › Calendriers and Réglages › Espace de travail. **Réglages › Diagnostic**: health « prête », and « Vérifier » (the database check) whenever you want.
6. After an update that adds tabs or columns, the Installation page comes back once: one click.

Advanced (optional): `npm run setup:sheets` and `npm run doctor:sheets` do the same from a terminal (`docs/DEPLOYMENT.md` §5).

2.9 Wise (bank, read-only)

What it's for: the Finances page and the Command Center's cash position: balances, statements, live transactions via webhook. The same read-only token also provides **Wise's USD→EUR rates** used for **supplier payments in dollars**. **The token can never move money.**

What you need: admin access to the **Wise Business** account, your phone (two-factor authentication).

Steps (Docs: https://docs.wise.com/guides/developer/auth-and-security/personal-api-token)

1. In the application: **Réglages › Intégrations › Wise › Configurer**. Keep the window open.
2. wise.com › Business account › **Settings › API tokens** (depending on the version: *Connect and manage apps › API tokens* or *Integrations and Tools › API tokens*) › **Add new token** › name "Valemont OS" › access **Read only** › confirm two-factor authentication. The token is shown **only once**: paste it straight into the field « Jeton d'API en lecture seule » (Read-only API token, `WISE_API_TOKEN`, shaped 8-4-4-4-12 letters and digits).
3. « Identifiant du profil entreprise » (Business profile id): leave empty, the app finds the Business profile itself.
4. (Optional, live transactions) At Wise, add a `balances#update` webhook with the **webhook address** shown in the form (`https://<your-app>/api/webhooks/wise`), in Wise › Settings › **Webhooks** if the menu exists on your account (Docs: https://docs.wise.com/guides/developer/webhooks/subscribe-to-webhooks). Then copy Wise's **production public key** for webhooks ("Event handling" page: https://docs.wise.com/guides/developer/webhooks/event-handling) and paste it **whole** (BEGIN and END lines included) into « Clé publique des webhooks Wise » (Wise webhook public key). No need to convert it: the app does it. If the Webhooks menu doesn't exist, skip this step: the **Synchroniser** button catches up on transactions.
5. **Enregistrer et tester**: the app reads your balances once. Green = connected.

Advanced (optional): **SCA key** to read statements when Wise requires strong authentication. It is made in a terminal (`openssl genrsa -out wise-private.pem 2048` , then `openssl rsa -pubout -in wise-private.pem -out wise-public.pem`); upload `wise-public.pem` in Wise › API tokens › **Manage public keys**, paste the content of `wise-private.pem` into « Clé privée pour les relevés (SCA) » (Private key for statements), then delete the files. In the EU/UK (PSD2), Wise may refuse statements through a personal token, even with the key: balances and the webhook still work, and the history comes from a **Wise statement CSV export** (Finances › « Importer un CSV Wise » (Import a Wise CSV)). Creating the webhook subscription through the API is also possible (see `docs/INTEGRATIONS.md`).

Where it goes: in the app (Configurer): `WISE_API_TOKEN` (required for Wise), `WISE_PROFILE_ID` , `WISE_WEBHOOK_PUBLIC_KEY_BASE64` , `WISE_SCA_PRIVATE_KEY_BASE64` (optional).

Check: green after **Enregistrer et tester**, then **Synchroniser** (statements). Balances appear in Finances. A credit automatically classified as revenue stays « à vérifier » (to review) until a founder has confirmed it.

Common problems: red 401 = token revoked or copied wrong (paste it again); "SCA required" = SCA key missing or rejected → CSV import; wrong profile = fill in the business profile id in the form.

2.10 Stripe (read-only)

What it's for: Stripe balance, sales, fees, payouts, MRR (Finances and Command Center). The key can neither charge nor refund. Stripe → Wise payouts are internal transfers: a sale is never counted twice.

Steps (Docs: https://docs.stripe.com/keys/restricted-api-keys, https://docs.stripe.com/webhooks)

1. In the application: **Réglages › Intégrations › Stripe › Configurer**.
2. Stripe Dashboard (**live** mode) › **Developers › API keys › Create restricted key** › name "Valemont OS". Set **Read** on: *Balance*, *Balance transactions*, *Subscriptions*, *Customers*. Everything else: **None**. Paste the key (`rk_live_…`) into « Clé restreinte (lecture seule) » (Restricted key, read-only). A `pk_` key is refused.
3. **Developers › Webhooks** (in Workbench: *Event destinations*) › **Add destination** › paste the **webhook address** shown in the form (`https://<your-app>/api/webhooks/stripe`) › events: `charge.*` , `payout.*` , `customer.subscription.*` , `invoice.paid` , `balance.available` .
4. Open the destination › **Signing secret** › Reveal (`whsec_…`) › paste it into « Secret de signature du webhook » (Webhook signing secret).
5. **Enregistrer et tester**: the app reads the balance once.

Where it goes: in the app (Configurer): `STRIPE_API_KEY` (required for Stripe), `STRIPE_WEBHOOK_SECRET` (optional but recommended).

Check: green after **Enregistrer et tester**, then **Synchroniser**; in Stripe, *Send test event* on the destination must answer 200.

Common problems: 403 "missing permission" = a resource isn't set to Read (the message says which one); 400 on the webhook = secret from another destination or from test mode.

2.11 Website: forms → signed webhook

What it's for: every form (contact, lead, booking) on the FR/ES websites creates an inbound lead (owner Théo for FR, Pablo for ES) and an « À répondre » item. An unsubscribed contact is blocked: nothing is created.

Steps

1. In the application: **Réglages › Intégrations › Formulaires du site › Configurer** › **Générer** next to « Secret de signature des formulaires » (Form signing secret). The secret (`WEBSITE_WEBHOOK_SECRET`) is saved encrypted and shown **only once**: copy it.
2. At the website host (Netlify › Site configuration › Environment variables), add `WEBSITE_WEBHOOK_SECRET` with that value and `VALEMONT_WEBHOOK_URL` with the **form webhook address** shown in the form (`https://<your-app>/api/webhooks/website`).
3. Netlify Forms: Netlify's outgoing webhook can't sign, so you need a small **submission-created** function (ready-to-copy code in `docs/INTEGRATIONS.md` › "Netlify Forms"). Docs: https://docs.netlify.com/functions/trigger-on-events/. Other host: any server-side code (Cloudflare Worker, Apps Script…) that signs the same way (`docs/INTEGRATIONS.md` › "Generic form handler"). This is the only "code" step: whoever manages the website does it once. **The secret must never be in browser code.**
4. In the form: honeypot field `bot-field` , hidden fields `lang` , `page` , UTM, `consent_version` , and a consent checkbox whose sentence is sent as `consent_text` .
5. **Enregistrer et tester**, then send a real test form.

Where it goes: in the app (Configurer): `WEBSITE_WEBHOOK_SECRET` ; on the website side (not in Valemont OS): the same value + `VALEMONT_WEBHOOK_URL` .

Check: green after **Enregistrer et tester** (the secret is present and long enough); the test form appears in Ma journée › À répondre (My Day › To answer).

Common problems: 401 `stale_timestamp` = server clock off by more than 5 min; 401 `bad_signature` = different secrets (after a new **Générer**, copy it to the host again) or signature computed on reformatted JSON (you must sign the exact text sent).

2.12 Messages to handle: Make / Zapier / n8n relay and Instagram

What it's for: gathering what's waiting for a reply (comments, DMs, LinkedIn notifications…) with a suggested reply in your language. **Nothing is ever sent automatically**: « Copier une réponse » (Copy a

reply) opens the conversation and you paste it yourself. Flagged email/WhatsApp replies and website leads already arrive without connecting anything.

Everything is set in **Réglages › Intégrations › Messages à traiter (To reply inbox) › Configurer**.

A. Relay (Make, Zapier or n8n)

1. **Générer** next to « Secret de signature du relais » (Relay signing secret, `INBOX_WEBHOOK_SECRET`): copy the value shown only once and paste it in the relay (variable / protected connection).
2. Trigger of your choice: "Instagram for Business › Watch comments", the relay's mailbox (Make *Mailhook*, *Email by Zapier*) fed by a **Gmail filter** that forwards LinkedIn notifications, etc.
3. Action: HTTP POST to the **relay address** shown in the form (`https://<your-app>/api/inbox/webhook`), JSON body (format in `docs/INBOX.md`), header `X-Valemont-Signature: sha256=<HMAC-SHA256 of the exact body with the secret>`: Make = function `sha256(body; "hex"; secret)` in HMAC mode; Zapier = *Code by Zapier* step; n8n = *Crypto* node (HMAC, SHA256, hex). `external_id` must be stable (it's the duplicate guard).

B. Instagram via the Meta app (see also §2.13 for the app itself)

1. In the form, **Générer** next to « Jeton de vérification Meta » (Meta verify token, `META_VERIFY_TOKEN`): it is already saved; copy it.
2. In the Meta app › **Webhooks** (or Instagram › *API setup* › *Configure webhooks*): callback URL = the **Meta address** from the form (`https://<your-app>/api/inbox/meta`), verify token = the copied value › **Verify and save**. (Docs: https://developers.facebook.com/docs/instagram-platform/webhooks)
3. Subscribe the `comments` and `messages` fields for both accounts (Valemont, 30/30).
4. **App settings › Basic › App secret** (32 letters and digits) › paste it into « Clé secrète de l'app Meta » (Meta app secret, `META_APP_SECRET`) › **Enregistrer et tester**.
5. (Optional) Name the accounts: setting `instagram_account_labels` (see `docs/INBOX.md`).
6. ⏳ Permissions `instagram_business_manage_comments`, `instagram_business_manage_messages` (or the equivalents `instagram_manage_comments` / `instagram_manage_messages`): to receive events from accounts that have no role in the app, Meta requires **App Review** (a few days to a few weeks, demo video, privacy policy). In the meantime, use the Make "Watch comments" relay. (Docs: https://developers.facebook.com/docs/resp-plat-initiatives/individual-processes/app-review)

Where it goes: in the app (Configurer): `INBOX_WEBHOOK_SECRET`, `META_VERIFY_TOKEN` (generated here), `META_APP_SECRET` (pasted from Meta).

Check: green after **Enregistrer et tester**; then a test comment on Instagram (or a call from the relay) appears in Ma journée › À répondre.

Common problems: Meta can't "verify and save" the URL = the token pasted at Meta differs from the generated one (generate it again and copy it again); 401 from the relay = signature computed on a

different text than the body sent, or the old secret still in the relay.

2.13 Instagram: statistics (read-only) and snapshot every 6 h

What it's for: followers, views, reach, profile visits, website clicks, and numbers for each post/Reel at 24 h, 72 h, 7 d, 30 d, stories every 6 h (Studio contenu, Croissance). Without a token: manual entry, marked « saisi à la main » (entered by hand).

What you need: each Instagram account as a **professional account** (Business or Creator) **linked to a Facebook Page** (Docs: https://developers.facebook.com/docs/instagram-platform/overview), all within **Valemont's Meta business portfolio**.

Steps (recommended method: Facebook Login + system user, non-expiring token)

1. In the application: **Réglages › Intégrations › Instagram › Configurer**.
2. https://developers.facebook.com › **My Apps › Create App** › type Business, attached to the Valemont portfolio (never a third party's personal account). Add the **Instagram** product (API with Facebook Login). (Docs: https://developers.facebook.com/docs/instagram-platform/instagram-api-with-facebook-login/)
3. **Business settings** (business.facebook.com/settings) › **Users › System users › Add** (admin role) › **Assign assets**: the app, the Page and the Instagram accounts. (Docs: https://developers.facebook.com/docs/marketing-api/system-users/)
4. **Generate token** › Valemont OS app › expiration **Never** › permissions `instagram_basic`, `instagram_manage_insights`, `pages_show_list`, `pages_read_engagement` › paste it into « Jeton d'accès » (Access token, `INSTAGRAM_ACCESS_TOKEN`).
5. Account ID: in the Graph API Explorer, `me/accounts?fields=instagram_business_account` → the number `instagram_business_account.id` › paste it into « Identifiant du compte Instagram » (Instagram account id, `INSTAGRAM_BUSINESS_ACCOUNT_ID`).
6. « Quel compte » (Which account): leave empty for `ig-valemont`, or enter `ig-thirty` for 30/30. « Date d'expiration du jeton » (Token expiry date): empty for a system user token.
7. **Enregistrer et tester**: the app reads the profile once. Standard Access on your own accounts: **no App Review needed** for statistics.

Advanced (optional): **two accounts at once** = variable `INSTAGRAM_ACCOUNTS_JSON` in Vercel (exact format in `.env.example`, then Redeploy), instead of the form. **Fallback method (Instagram Login, 60-day token)**: permissions `instagram_business_basic` + `instagram_business_manage_insights`, Vercel variable `INSTAGRAM_API_BASE_URL=https://graph.instagram.com/v21.0`, and the expiry date in the form. The application shows the days remaining and creates a task 14 days before; renewal: https://developers.facebook.com/docs/instagram-platform/reference/refresh_access_token, then paste the new token in **Configurer** (no redeploy).

Snapshot every 6 h: already scheduled by Vercel Cron (`CRON_SECRET` , Phase 2). The row **Capture planifiée (Snapshot job) › Configurer** is only needed if another scheduler calls it (Apps Script, GitHub Actions, cron-job.org): **Générer** the « Jeton de la tâche » (Job token), copy it once into the scheduler, which calls the **job address** shown every 6 hours with the header `Authorization: Bearer <token>` (or generate the job signing secret instead if the scheduler can sign, see `docs/INTEGRATIONS.md`) › **Enregistrer** (Save).

Where it goes: in the app (Configurer): `INSTAGRAM_ACCESS_TOKEN` , `INSTAGRAM_BUSINESS_ACCOUNT_ID` , `INSTAGRAM_ACCOUNT_KEY` , `INSTAGRAM_TOKEN_EXPIRES_AT` , `SNAPSHOT_JOB_TOKEN` , `SNAPSHOT_JOB_SECRET` ; Vercel (advanced): `INSTAGRAM_ACCOUNTS_JSON` , `INSTAGRAM_API_BASE_URL` .

Check: green after **Enregistrer et tester**, then **Synchroniser**; the « Capture planifiée » row shows the last run; Studio contenu shows the numbers.

Common problems: no `instagram_business_account` = account not professional or not linked to the Page; story statistics "not available" under 5 viewers (this is normal, never 0); demographics missing under 100 followers.

2.14 LinkedIn (imports, and API later)

What it's for: impressions, followers and countries for the Valemont / 30/30 Pages and your creator profiles, in Croissance. LinkedIn prospecting stays manual (log + opening the profile): no scraping, no automatic messages.

Now: the exports

1. LinkedIn Page (as admin) › **Analytics** › tab (Content, Visitors, Followers) › **Export** › period › **.xls** file.
2. Creator profile: your profile's **Analytics** › **Export** › **.xlsx** file.
3. In the application: Croissance › **Ajouter des chiffres** (Add numbers) › **Importer un export LinkedIn** (Import a LinkedIn export). Re-importing the same file adds no duplicates. Once a week is enough.

Later (⏳ optional): Community Management API

- Create a **new** dedicated LinkedIn Developers app (no other product), linked to Valemont's **verified Page**, request *Development tier* access, then *Standard tier* with a video of each use case. Long review, possible refusal (you then need a new app). The adapter will only be built after approval. (Docs: https://learn.microsoft.com/en-us/linkedin/marketing/community-management/community-management-overview, https://learn.microsoft.com/en-us/linkedin/marketing/increasing-access)

Where it goes: nothing to paste. **Expected status:** *En attente d'approbation* (normal).

2.15 GA4 and Search Console (CSV imports)

What it's for: website visits and leads by source (including AI engines) in Croissance; queries and pages (clicks, impressions, position) in Croissance › SEO and the Studio contenu optimizer.

GA4 (Docs: https://support.google.com/analytics/answer/9317657)

1. analytics.google.com › **Reports** › Acquisition › Traffic acquisition › period (e.g. last 7 days).
2. **Share this report** (top right) › **Download File** › **CSV**.
3. Croissance › **Ajouter des chiffres** › import the GA4 CSV.

Search Console (Docs: https://support.google.com/webmasters/answer/7576553)

1. search.google.com/search-console › website property › **Performance** › period.
2. **Export** › **Download CSV**.
3. Croissance › SEO › **Ajouter des chiffres Search Console** (Add Search Console numbers).

Where it goes: nothing to paste. **Expected status:** *Import uniquement* with the date of the last import. Suggested routine: on Monday, both exports.

2.16 Mon tracker (My tracker: your personal tracking app)

What it's for: each of your activities (calls, emails, visits…) is sent to your personal tracker, signed; your tracker can also read your activities through a feed. LIVE only. Each of you configures **your own**, signed in with your own account: the other founder neither sees nor changes your values.

Steps (in **Réglages › Intégrations › Mon tracker › Configurer**)

1. In your tracking app, create an **incoming webhook** that accepts JSON and copy its address (https required) › paste it into « Adresse du tracker » (Tracker address, `TRACKER_WEBHOOK_URL_<name>`).
2. **Générer** next to « Secret de signature » (Signing secret, `TRACKER_WEBHOOK_SECRET_<name>`): copy the value shown only once and give it to your tracker so it can verify the `X-Valemont-Signature` header (verification code in `docs/TRACKER.md`).
3. (Optional) Feed: **Générer** the « Jeton du flux » (Feed token, `TRACKER_FEED_TOKEN_<name>`); your tracker calls the **feed address** shown (`https://<your-app>/api/tracker/feed?from=YYYY-MM-DD&to=YYYY-MM-DD`) with `Authorization: Bearer <token>`.
4. **Enregistrer et tester**: one signed test event goes to your tracker.

Advanced (optional): `TRACKER_DAILY_SUMMARY_THEO=1` (or `_PABLO`) in Vercel to also send the day's summary when the daily review is saved.

Where it goes: in the app (Configurer, each your own): `TRACKER_WEBHOOK_URL_*`, `TRACKER_WEBHOOK_SECRET_*`, `TRACKER_FEED_TOKEN_*`; Vercel (advanced): `TRACKER_DAILY_SUMMARY_*`.

Check: green after **Enregistrer et tester** (or Réglages › **Mon tracker** › **Envoyer un événement de test** (Send a test event)); « Envoyer cette période » (Send this period) fills a gap without duplicates.

Common problems: `http://` address refused; the tracker must answer 2xx in under 5 s.

2.17 Backups, scheduled jobs, restore

What it's for: a daily Google Sheet copy, verified, kept 30 days; a tested restore.

Steps

1. Backup folder:
   - **Google Workspace**: create a **shared drive** "Valemont OS – sauvegardes", add the service account as **Content manager**;
   - **personal Google account**: a service account has no storage space. Create a normal folder, share it as Editor, and make the copy with your own small **Apps Script** (copy-paste code and daily trigger in `docs/DEPLOYMENT.md` › 6. Backups, all in the browser).
2. In the application: **Réglages › Intégrations › Sauvegardes › Configurer** › paste the **folder's address** (or only its ID) into « Dossier des sauvegardes » (Backups folder, `BACKUP_DRIVE_FOLDER_ID`). « Jours à conserver » (Days to keep): empty = 30 (1 to 365).
3. Scheduling: nothing to do, Vercel Cron runs the backup at 02:17 UTC with `CRON_SECRET` (Phase 2). Without a scheduler, the first visit of the day starts the copy. (Optional) For another scheduler: **Générer** the « Jeton de la tâche » (Job token) and copy it once into it.
4. **Enregistrer et tester**: the app lists the folder once.
5. **Restore drill** (the next day, then every quarter), no terminal: Réglages › Diagnostic › **Sauvegardes** › « Vérifier » (Verify) on the day's copy (it is compared with the current database: tabs, headers, row counts). For a real need: open the copy › File › **Make a copy**, share it with the service account as Editor, put its ID in `SHEETS_SPREADSHEET_ID` (Vercel) › Redeploy; the Installation page checks everything. **Don't switch production over** for a drill, and keep the old Sheet untouched.
6. (Optional) An uptime monitor (UptimeRobot, Better Stack…) on `https://<your-app>/api/health`, at most once a minute.

Advanced (optional): full restore from a terminal into a **new** empty Sheet: `npm run restore:sheets -- --from-sheet <backup copy id> --to <new Sheet id>`, then `SHEETS_SPREADSHEET_ID=<new id> npm run doctor:sheets` (the script refuses the production Sheet and any non-empty Sheet; details in `docs/DEPLOYMENT.md` §7). `BACKUP_JOB_SECRET` (signing) goes in Vercel.

Where it goes: in the app (Configurer): `BACKUP_DRIVE_FOLDER_ID`, `BACKUP_RETENTION_DAYS`, `BACKUP_JOB_TOKEN`; Vercel: `CRON_SECRET` (Phase 2), `BACKUP_JOB_SECRET` (advanced).

Check: Réglages › Diagnostic › **Sauvegardes** (Backups): "Valemont OS backup <today's date>" present and **vérifiée** (verified); « Vérifier » (Verify) button to compare with the current database.

Common problems: `storageQuotaExceeded` = personal account without a shared drive → Apps Script solution; red on test = folder not shared with the service account.

2.18 Final checks (Phase 10)

1. Réglages › Intégrations: click **Tester la connexion** on every configured row → **Connecté**.
2. Réglages › Diagnostic: health « prête », backup verified.
3. Go through the "Stage 4 gate: live validation" checklist in `docs/ACCEPTANCE.md` together (both sign-ins, a call seen by the other founder in under a minute, edit conflict, `+33…` intact in the Sheet, calendar, files, LinkedIn imports, backup, restore, `/api/health`, security headers).
4. Record the dated result in `docs/STATUS.md`. An integration is only **VERIFIED_LIVE** after this real test.

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3. All the variables

Two places:

- **Vercel (base)**: **Vercel › Settings › Environment Variables** (Production, *Sensitive* for secrets), then **Redeploy**. These are the settings the app needs before it can open the Sheet; Réglages › Intégrations shows them read-only.
- **In the app (Configurer)**: **Réglages › Intégrations › Configurer**; applied immediately, no redeploy. A variable with the same name in Vercel is still possible and **takes priority**.

"You" = a value you generate yourselves (§2.1); "Générer" = the form's Generate button.

Variable	Where it goes	Required?	Where to find it	Integration
`APP_MODE`	Vercel (base)	Required (`live`)	You	Core
`SESSION_SECRET`	Vercel (base)	Required	You (32+ chars, `/setup` › Generate a secret)	Core
`CONNECTORS_ENCRYPTION_KEY`	Vercel (base)	Strongly recommended	`/setup` › Generate a secret	Encryption of in-app keys
`CONNECTORS_ENCRYPTION_KEY_PREVIOUS`	Vercel (base)	Only while changing the key	The old key	Encryption of in-app keys
`APP_BASE_URL`	Vercel (base)	Optional on Vercel (automatic), required elsewhere	Your domain, `https://…` with no trailing `/`	Core, Sign-In, webhooks
`SYNC_POLL_SECONDS`	Vercel (base)	Optional (30)	30 to 60	Core
`LOG_LEVEL`	Vercel (base)	Optional (`info`)	`debug` / `info` / `warn` / `error` / `silent`	Logs
`FOUNDER_THEO_EMAIL` , `FOUNDER_PABLO_EMAIL`	Vercel (base)	Required	Your Google addresses	Sign-In
`FOUNDER_THEO_TIMEZONE` , `FOUNDER_PABLO_TIMEZONE`	Vercel (base)	Optional (default `Europe/Paris` , `Europe/Madrid`)	`Europe/Paris` , `Europe/Madrid`	Core
`GOOGLE_OAUTH_CLIENT_ID` , `GOOGLE_OAUTH_CLIENT_SECRET`	Vercel (base)	Required	Google Auth Platform › Clients	Sign-In, Drive, Gmail
`SHEETS_SPREADSHEET_ID`	Vercel (base)	Required	Sheet URL	Google Sheets
`GOOGLE_SERVICE_ACCOUNT_KEY_BASE64`	Vercel (base)	Required	Service account JSON key, pasted as-is (or in base64)	Sheets, Drive, Calendar, backups
`PRIVATE_SHEET_ID_THEO` , `PRIVATE_SHEET_ID_PABLO`	Vercel (base)	Optional	URL of each founder's private Sheet	Google Sheets (private)
`CRON_SECRET`	Vercel (base)	Recommended on Vercel	`/setup` › Generate a secret (24+ chars)	Backups + snapshot
`WISE_API_TOKEN`	In the app (Configurer)	Required for Wise	Wise › Settings › API tokens (Read only)	Wise
`WISE_PROFILE_ID`	In the app (Configurer)	Optional (found automatically)	BUSINESS profile	Wise

Variable	Where it goes	Required?	Where to find it	Integration
`WISE_WEBHOOK_PUBLIC_KEY_BASE64`	In the app (Configurer)	Optional	Wise webhooks production public key, pasted whole	Wise
`WISE_SCA_PRIVATE_KEY_BASE64`	In the app (Configurer)	Optional (advanced)	Your private RSA key (public part uploaded to Wise), pasted whole	Wise
`STRIPE_API_KEY`	In the app (Configurer)	Required for Stripe	Stripe › Developers › API keys › restricted key `rk_live_…`	Stripe
`STRIPE_WEBHOOK_SECRET`	In the app (Configurer)	Optional (recommended)	Stripe › Webhooks › destination › secret `whsec_…`	Stripe
`WEBSITE_WEBHOOK_SECRET`	In the app (Configurer)	Required for forms	Générer, then also on the website side	Website forms
`INBOX_WEBHOOK_SECRET`	In the app (Configurer)	Required for the relay	Générer, then also in Make/Zapier/n8n	Messages to handle
`META_APP_SECRET`	In the app (Configurer)	Optional	Meta app › Settings › Basic	Messages (Instagram)
`META_VERIFY_TOKEN`	In the app (Configurer)	Optional	Générer, then also in the Meta Webhooks config	Messages (Instagram)
`INSTAGRAM_ACCESS_TOKEN`	In the app (Configurer)	Required for 1 account	Meta system user (or Instagram Login)	Instagram
`INSTAGRAM_BUSINESS_ACCOUNT_ID`	In the app (Configurer)	Required for 1 account	Graph API Explorer	Instagram
`INSTAGRAM_ACCOUNT_KEY`	In the app (Configurer)	Optional	`ig-valemont` (default) or `ig-thirty`	Instagram
`INSTAGRAM_TOKEN_EXPIRES_AT`	In the app (Configurer)	Optional	Expiry date of a 60-day token	Instagram
`INSTAGRAM_ACCOUNTS_JSON`	Vercel (advanced)	Optional (instead of the 4 above)	You (JSON list, see `.env.example`)	Instagram
`INSTAGRAM_API_BASE_URL`	Vercel (advanced)	Optional	Only `https://graph.instagram.com/v21.0` with Instagram Login	Instagram
`SNAPSHOT_JOB_TOKEN`, `SNAPSHOT_JOB_SECRET`	In the app (Configurer)	Optional	Générer (if a scheduler other than Vercel)	Snapshot job

Variable	Where it goes	Required?	Where to find it	Integration
`TRACKER_WEBHOOK_URL_THEO` / `_PABLO`	In the app (Configurer, each your own)	Optional	Your tracking app (https)	My tracker
`TRACKER_WEBHOOK_SECRET_THEO` / `_PABLO`	In the app (Configurer, each your own)	Required with the address	Générer	My tracker
`TRACKER_FEED_TOKEN_THEO` / `_PABLO`	In the app (Configurer, each your own)	Optional	Générer	My tracker
`TRACKER_DAILY_SUMMARY_THEO` / `_PABLO`	Vercel (advanced)	Optional	`1` to enable	My tracker
`BACKUP_DRIVE_FOLDER_ID`	In the app (Configurer)	Optional (recommended)	Address of the backup folder / shared drive	Backups
`BACKUP_RETENTION_DAYS`	In the app (Configurer)	Optional (30)	1 to 365	Backups
`BACKUP_JOB_TOKEN`	In the app (Configurer)	Optional	Générer (if a scheduler other than Vercel)	Backups
`BACKUP_JOB_SECRET`	Vercel (advanced)	Optional	You (if a scheduler that signs)	Backups
`SHEETS_API_BASE_URL`, `GOOGLE_API_BASE_URL`, `WISE_API_BASE_URL`, `STRIPE_API_BASE_URL`, `DEMO_BACKEND`, `FAKE_SHEETS_URL`, `FAKE_SHEETS_SPREADSHEET_ID`, `DEMO_ACCESS_CODE`	**Never in production**	—	Tests and demo only	—

On the website side (not in Valemont OS): `WEBSITE_WEBHOOK_SECRET` (same value) and `VALEMONT_WEBHOOK_URL`. Calendar IDs and the Drive folder ID are entered **in the application** (Réglages › Calendriers, Réglages › Espace de travail), not as variables.

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4. Security reminders

- **Keys pasted in the application are encrypted** (AES-256-GCM) before they are saved: the Sheet holds only unreadable text (`IntegrationSecrets` tab), and the encryption key (`CONNECTORS_ENCRYPTION_KEY`) stays in Vercel, never in the Sheet. Non-secret identifiers (Wise profile, Instagram account, backup folder) are saved in plain text in the settings.
- **A saved key is never shown again**: only « Configuré · se termine par 1234 » (Set · ends with 1234), who saved it and when. To change it, paste a new value or click **Générer** again; **Supprimer** (Delete) erases it. A generated secret is shown only once: copy it straight away into the other service.
- **Only Théo and Pablo can change the keys** (signed in with their founder Google account). Every change is written to the audit log (the field name only, never the value). Tracker values belong to each founder.
- **Base settings live only in Vercel** (ticked *Sensitive*) and in your password manager. No secret in a chat, an email, directly in the Sheet, a ticket, a screenshot or the code. Logs mask keys, emails and phone numbers.
- **Least privilege everywhere:** service account with no role (only the shares it needs), Drive folder as Viewer, personal calendars as "free/busy only", Wise **read-only**, Stripe **restricted Read key**, Instagram statistics/read, Gmail **send only**, Drive as `drive.file` (only the files created by the app).
- **What the application never does:**
  - send an email, a WhatsApp, an Instagram or LinkedIn reply without **your click**;
  - read your Gmail inbox;
  - move money, pay, refund (Wise and Stripe are read-only);
  - contact an unsubscribed contact (blocked on the server, for both of you);
  - show one founder the other's private data (private tasks, personal calendar details);
  - use the demo instead of the real data: without the base settings, LIVE mode shows a configuration error (in the demo, nothing you paste is really saved).
- **Rotation:**
  - leak or doubt about a service key → regenerate it at the provider, then **Configurer** › paste the new value › **Enregistrer et tester** (no redeploy); for a secret we create, **Générer** again and copy it into the other service again;
  - Google key → new key, Vercel variable, Redeploy, **then** delete the old one;
  - changing `SESSION_SECRET` = everyone is signed out (this is how you revoke a stolen session). Without `CONNECTORS_ENCRYPTION_KEY` , the keys saved in the app also become unreadable ("unreadable, paste it again"): that's why this key is set in Phase 2;
  - changing `CONNECTORS_ENCRYPTION_KEY` : set the new value, the old one in `CONNECTORS_ENCRYPTION_KEY_PREVIOUS` , Redeploy, then Réglages › Intégrations › «

    Rechiffrer avec la nouvelle clé » (Re-encrypt with the new key); only then remove the old one (details: `docs/SECURITY.md` §12b);
    - 60-day Instagram token: the application creates a task 14 days before expiry;
    - a contractor leaves or you change computer: redo the tokens they may have seen.
- **One owner per resource, and it's Valemont:** Google accounts, Meta portfolio, Vercel account, Wise, Stripe. Turn on two-factor authentication everywhere.
- **Backups:** the downloadable JSON copy contains personal data, keep it somewhere private. A restore always goes into a **new** Sheet, never on top of production.

Delays to plan for (summary)

What	Delay	Meanwhile
Google verification for `gmail.send` (External app in production)	several days to a few weeks	Internal app (Workspace), or Testing mode with your two addresses, or pre-filled Gmail draft
Meta App Review (Instagram comments/messages)	a few days to a few weeks	Make "Watch comments" relay to Messages à traiter
LinkedIn Community Management API	long, not guaranteed	.xls/.xlsx exports imported every week
Vercel domain certificate	a few minutes to a few hours (DNS)	`*.vercel.app` address for testing (remember to add the new Google URIs afterwards)